

Invariant Coordinate Selection (ICS) - Julia tutorial

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Introduction

Invariant Coordinate Selection (ICS) is an unsupervised multivariate statistical method used for dimension reduction, outlier detection, and cluster identification. By simultaneously diagonalizing two different scatter matrices, ICS projects high-dimensional data into a new, meaningful coordinate system to easily uncover underlying structures. It is particularly useful for dimension reduction. Unlike PCA, ICS is not based on variance maximization but on the maximization/minimization of a generalized kurtosis, and it is invariant not only to orthogonal data transformations but also to any affine transformation.

The **Julia** `ICSTools` package brings the main functionalities of the **R** `ICS` package to **Julia**, offering tools for identifying and selecting invariant coordinates in multivariate data. It includes various covariance estimators, transformation settings, and plotting utilities. Our extensive testing ensures results consistent with the **R** package, making it easy for users to transition from **R** to **Julia** or start fresh with ICS.

The source code of the **Julia** package `ICSTools` is available at our GitHub Repository: `ICSTools`.

The complete documentation of the API with examples can be found at `ICSTools` Documentation.

The results of the test coverage are available at `ICSTools` Tests.

To run the code from this tutorial you will need to install **Julia**. The recommended way of installing **Julia** is based on `juliaup` which (on Windows) can be found here. Further you will need an IDE for more comfortable running of the code - the preferred one is VSCode with installed **Julia** plugin which can be obtained from here.

The **R** `Markdown` file of the tutorial could be run in **RStudio** but you will need also **R** installed.

In this GitHub repository is available also a Pluto notebook containing a part of the tutorial.

All the code chunks of the repository are extracted into a Julia script.

```
import Pkg;
Pkg.add("Robustbase"); using robustbase # for MCD
Pkg.add("ICSTools"); using ICSTools
Pkg.add("Plots"); using Plots
Pkg.add("DataFrames"); using DataFrames
Pkg.add("GLMakie"); using GLMakie # for pairplots() and related
Pkg.add("PairPlots"); using PairPlots # for pairplots() and related
Pkg.add("RCall"); using RCall # To access R data sets; to perform tests
Pkg.add("Clustering"); using Clustering # for kmeans() and randindex()
Pkg.add("Test"); using Test # for testing against R
Pkg.add("BenchmarkTools"); using BenchmarkTools # for benchmarking scatter functions
Pkg.add("Random"); using Random # for testing and benchmarking
```

Getting started

```
using ICSTools

# Load dataset
using Robustbase
X=wood[:,1:5];

## Instantiate ICSModel object with all default parameters
ics = ICSTools.ICSTools();
show(ics)

##
## ICS based on two scatter matrices
## S1: cov2
## S1_args: Dict{Symbol, Any}()
## S2: covW
## S2_args: Dict{Symbol, Any}()
##
## Information on the algorithm:
## algorithm: whiten
## center: false
## fix_signs: scores
##
## Model is not fitted yet!

## Alternative instantiations:

## 1. With values for S1 and S2
ics1 = ICSTools.ICSTools(S1=cov2, S2=cov4);

## 2. With arguments for S2
ics2 = ICSTools.ICSTools(S1=cov2, S2=covW, S2_args=Dict{Symbol, Any}(:alpha=>1, :cf=>2));

## 3. With algorithm
ics3 = ICSTools.ICSTools(S1=cov2, S2=covW, algorithm="standard");
```

With the function `fit_predict!()` we will fit the ICS model and transform the data - equivalent to the function `ICS-S3()` from the R package `ICS`. Alternatively, this can be done in two separate steps: `fit!()` and `predict!()`, as is usually done in machine learning pipelines.

```
scores = fit_predict!(ics, X);
show(ics)
```

```
##
## ICS based on two scatter matrices
## S1: cov2
## S1_args: Dict{Symbol, Any}()
## S2: covW
## S2_args: Dict{Symbol, Any}()
##
## Information on the algorithm:
## algorithm: whiten
## center: false
## fix_signs: scores
##
## The generalized kurtosis measures of the components are:
## IC_1: 6.6248
## IC_2: 6.1884
## IC_3: 5.2813
## IC_4: 4.8105
## IC_5: 3.9477
##
## The coefficient matrix of the linear transformation is:
##
```

	x1	x2	x3	x4	x5
## IC_1	1.19971	-31.83104	2.33736	-18.83172	-5.98021
## IC_2	-19.90580	58.80711	7.75780	19.47824	-28.44422
## IC_3	5.15339	11.03716	13.29070	-4.88970	5.98969
## IC_4	-2.44957	-43.70081	12.92686	-9.08586	21.63247
## IC_5	-4.00962	35.25209	-0.55332	-2.70552	-1.55875

```
##
scores
```

```
## 20x5 Matrix{Float64}:
## -16.7574 -15.0133 12.7086 13.2842 -1.58809
## -17.8712 -15.5105 14.5037 13.5282 -0.9788
## -17.4835 -16.7649 14.0566 14.5065 -1.05921
## -17.3958 -15.86 13.8091 15.3582 0.918907
## -16.961 -16.0117 14.0718 15.6424 -1.31639
## -17.2711 -15.9198 13.6709 14.8956 1.07561
## -15.5143 -12.7103 14.0587 14.3786 -0.447561
## -17.7991 -14.5827 13.2857 14.3302 1.32258
## -15.5807 -14.3794 14.7813 15.4326 -0.89646
## -18.7687 -14.5552 16.3595 14.2917 -0.53292
## -17.3183 -15.2455 14.7408 12.3598 0.00287171
## -16.1634 -15.786 14.4912 12.3209 -0.97496
## -17.3042 -15.2093 16.0166 14.9641 -0.841011
## -17.6189 -15.4777 13.5234 14.3221 -1.61633
## -17.8304 -14.0505 13.4756 14.4841 -1.32798
## -19.4147 -12.9501 13.376 13.7981 -0.811453
## -19.1585 -15.2576 14.3655 15.2923 -1.35671
## -17.0606 -14.5333 13.2685 16.1647 -1.33691
## -17.6047 -15.0584 13.2403 14.2924 1.399
## -17.8348 -15.8362 12.1999 14.0898 -1.57377
```

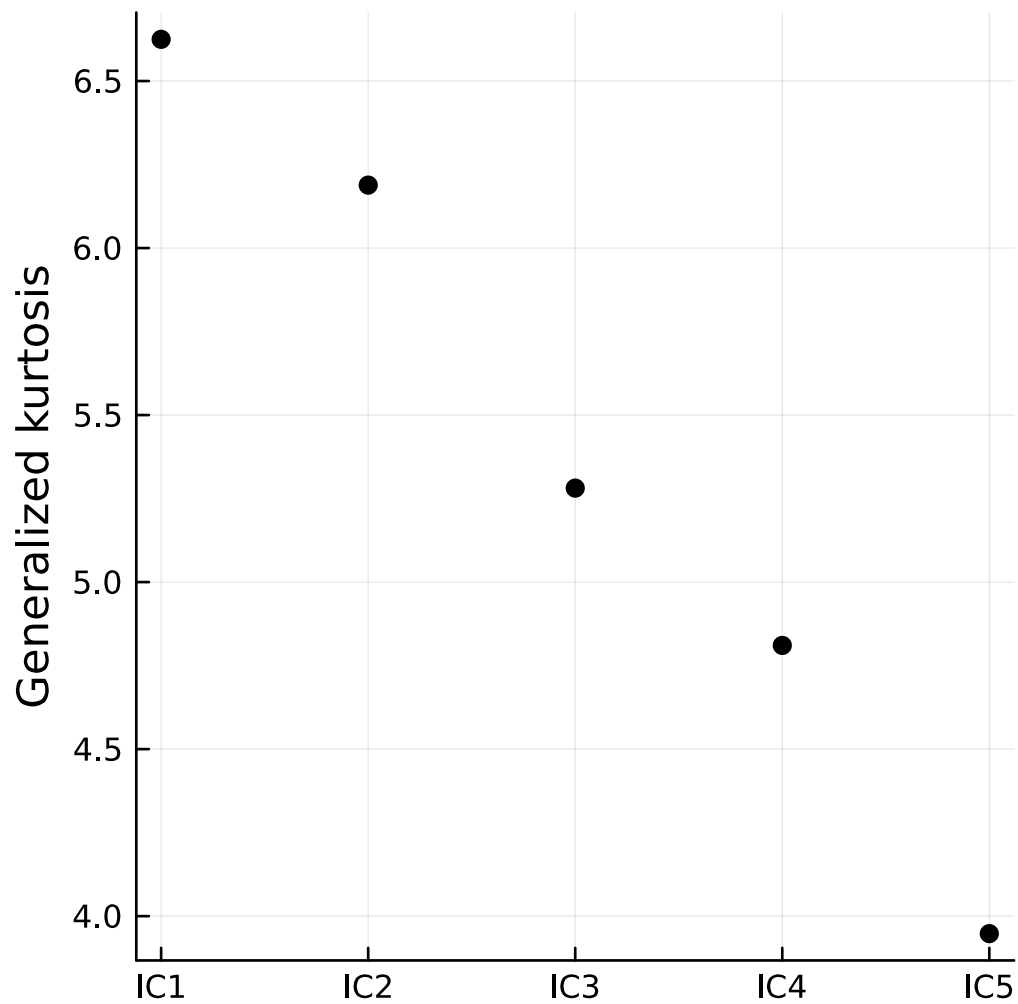
The following example illustrates how to plot the transformed/predicted data (invariant components) using

`component_plot2()` and the kurtosis of the invariant components (corresponding to the eigenvalues of the joint diagonalization problem) using `scree_plot()`.

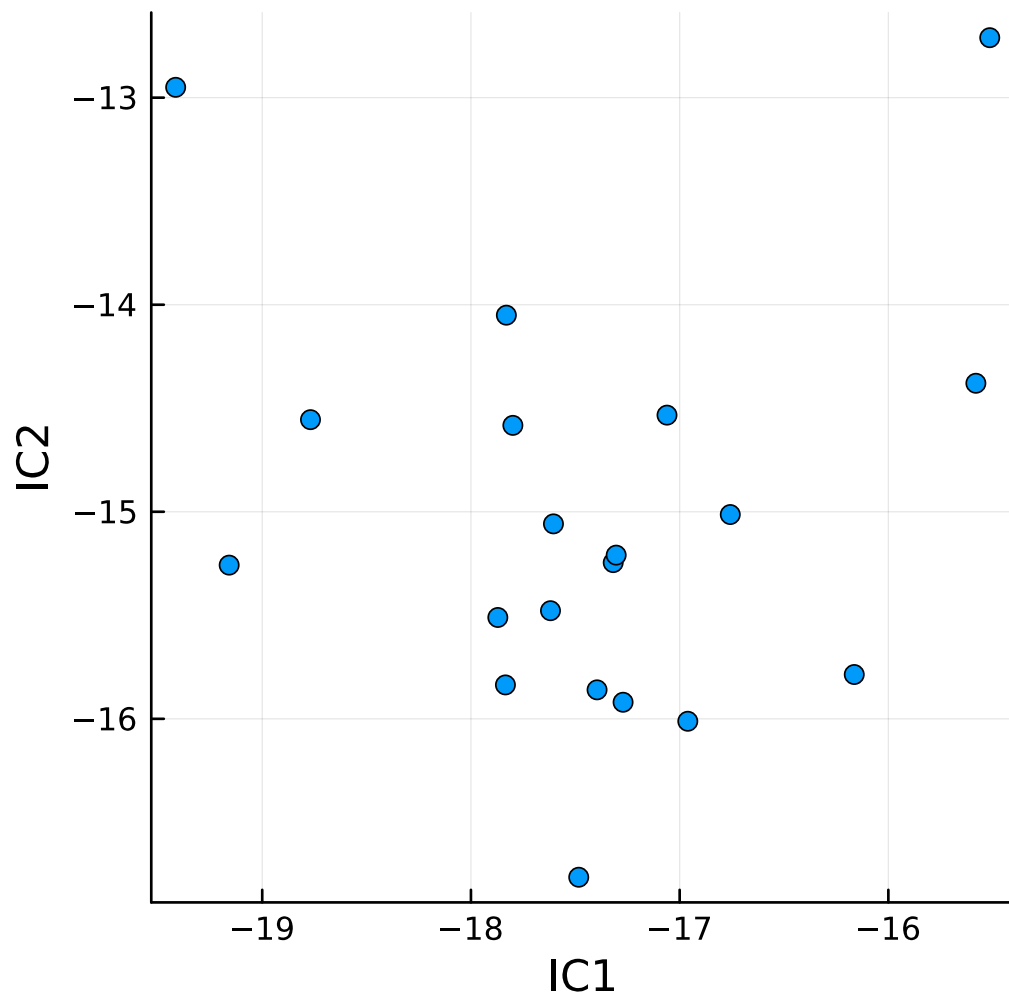
```
## scree plot and 2-dimensional component plot  
using Plots  
gr()
```

```
## Plots.GRBackend()
```

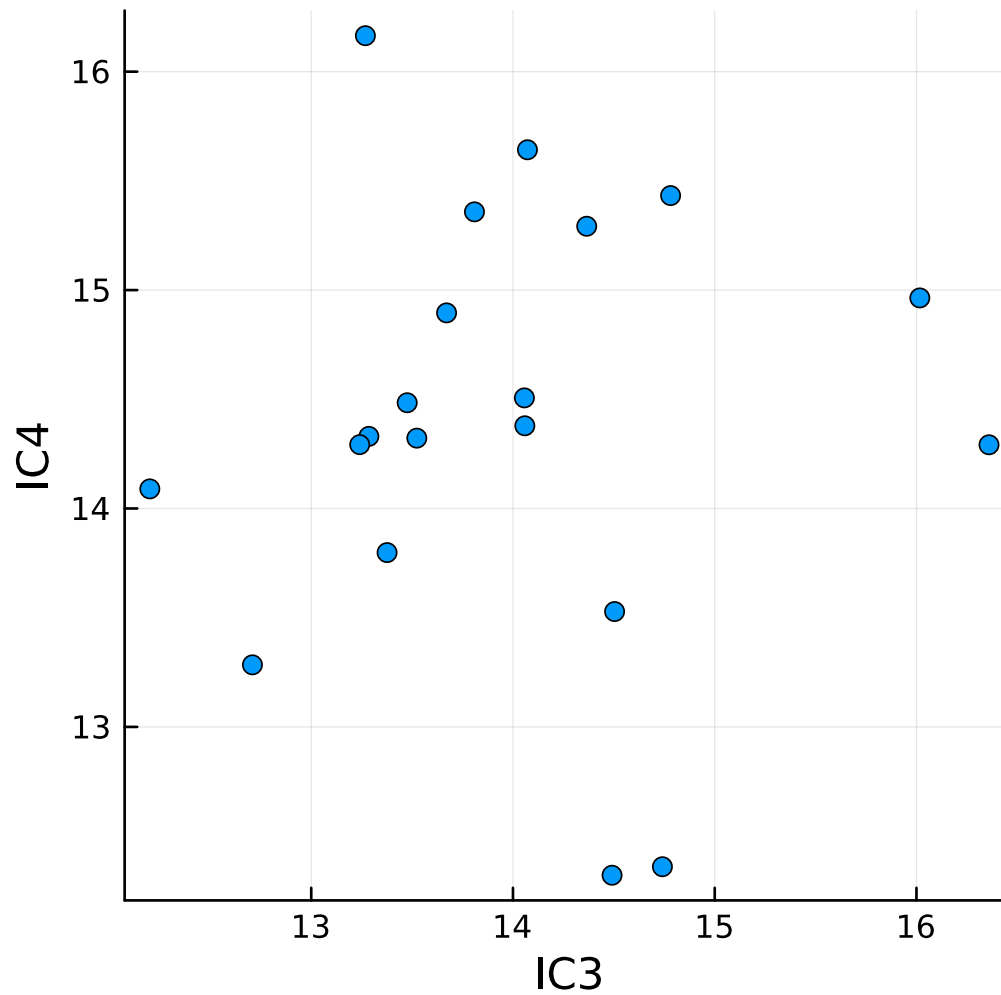
```
scree_plot(ics)
```



```
component_plot2(ics) # by default the first two components
```



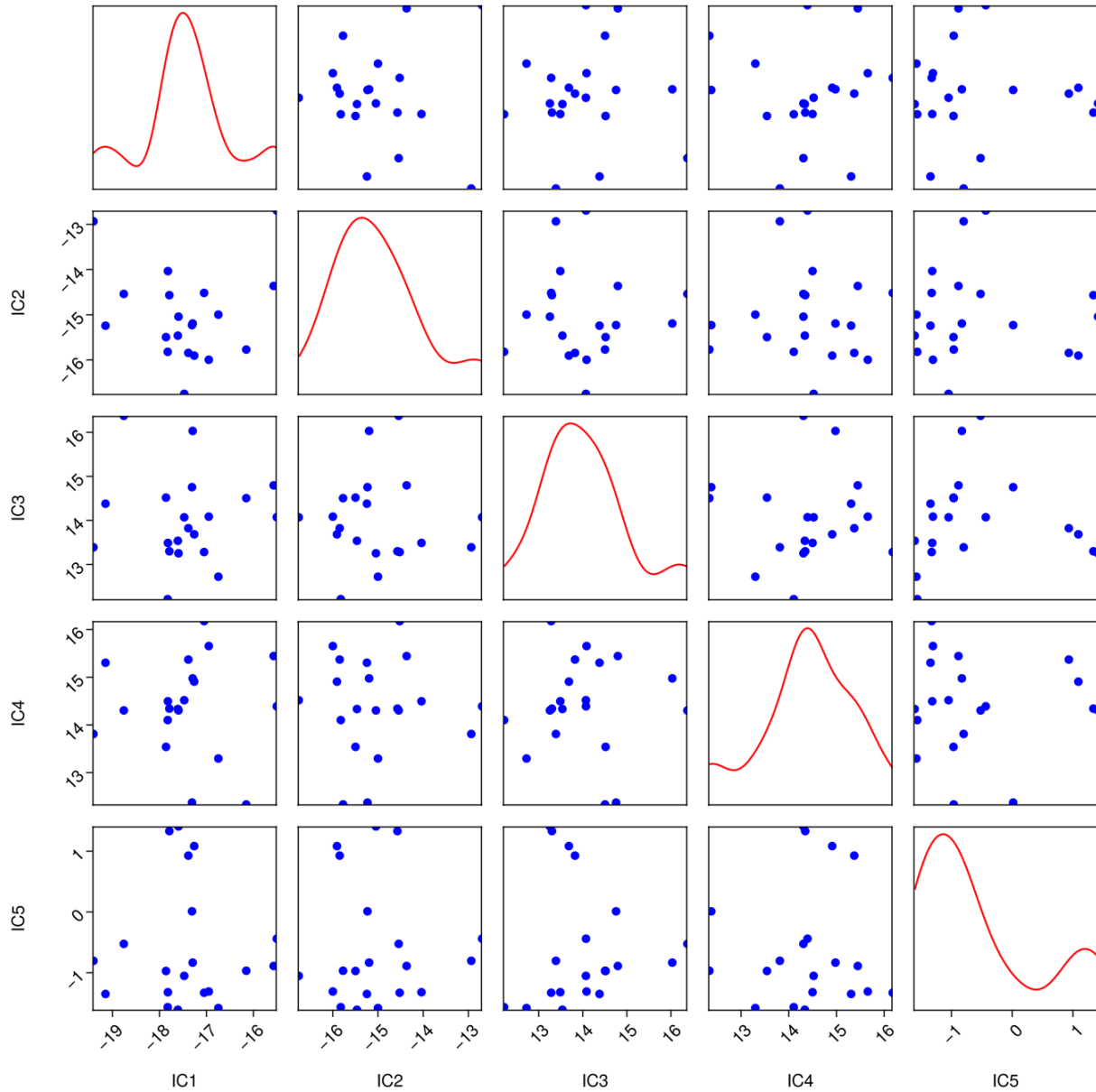
```
component_plot2(ics, select=[3,4]) # select components 3 and 4
```



```
using DataFrames
using GLMakie
using PairPlots
ss = DataFrame(scores, ["IC$i" for i=1:size(scores,2)]);
fig = pairplot(ss, fullgrid=true);

fig = pairplot(
    ss => (PairPlots.Scatter(markersize=10, color=:blue),
        PairPlots.MarginDensity(color=:red)),
    fullgrid=true
)

save("images/wood-scores.png", fig)
```



The scatters

ICS is based on simultaneously diagonalizing two different scatter matrices. The ICSTools package contains a number of implementations of scatter matrices, almost all scatters that are available in the different packages in R. Each scatter function takes a data matrix or a data frame and returns a structure with the following fields:

- (optional) location
- scatter matrix
- label

```
using ICSTools
```

```
# Load dataset
```

```
using Robustbase
X=wood[:,1:5];
```

```
cov4(X)
```

```
## -> Scatter: COV4
## Location:
## [0.55085, 0.13304, 0.50865, 0.5112, 0.90695]
##
## Scatter:
## 5×5 Matrix{Float64}:
##  0.00571919  -4.00121e-5   0.00240916   0.00157329  -0.00172161
## -4.00121e-5   0.000296558  -0.000155744  -0.000411835   0.000377114
##  0.00240916  -0.000155744   0.00302313   0.000877546  -0.00117088
##  0.00157329  -0.000411835   0.000877546   0.00274575  -0.000104867
## -0.00172161  0.000377114  -0.00117088  -0.000104867   0.0020241
```

Some scatter functions can take optional arguments. All functions are implemented in the ICSTools package, `mcd_raw()` and `mcd_rew()` use the Julia package Robustbase for the implementation of the FASTMCD algorithm.

```
covW(X, alpha=2, cf=4)
```

```
## -> Scatter: COVW
## Location:
## [0.55085, 0.13304, 0.50865, 0.5112, 0.90695]
##
## Scatter:
## 5×5 Matrix{Float64}:
##  0.90569      0.0204782   0.352329      0.210905   -0.211631
##  0.0204782    0.0402795  -0.00954821  -0.0386719   0.0530409
##  0.352329    -0.00954821   0.484683      0.0984048  -0.186089
##  0.210905    -0.0386719   0.0984048     0.463618   0.0586492
## -0.211631    0.0530409  -0.186089     0.0586492   0.341285
```

When scatter functions are used as arguments to `ICSModel()`, the optional arguments can be passed as data dictionaries:

```
ics1 = ICSModel(S1=cov2, S2=covW, S2_args=Dict{Symbol, Any}(:alpha=>2, :cf=>4));
ics2 = ICSModel(S1=mcd_raw, S2=cov2, S1_args=Dict{Symbol, Any}(:nsamp=>1000, :alpha=>0.75));
```

Table 1. List of scatter functions implemented in the Julia package ICSTool.

	Function	Location	Optional parameters
Covariance	<code>cov2</code>	Yes	-
Fourth-moment covariance	<code>cov4</code>	Yes	-
One-step M-estimator	<code>covW</code>	Yes	<code>alpha</code> , <code>cf</code>
One-step Tyler shape matrix	<code>covAxis</code>	Yes	-
Multivariate t-distribution estimator	<code>tM</code>	Yes	<code>alg</code> , <code>mu_init</code> , <code>V_init</code> , <code>gamma_init</code> , <code>eps</code> , <code>maxiter</code>

	Function	Location	Optional parameters
Cauchy location and scatter	mlc	Yes	alg, mu_init, V_init, gamma_init, eps, maxiter
Raw MCD	mcd_raw	Yes	alpha, nsamp
Rewighted MCD	mcd_rwt	Yes	alpha, nsamp
Pairwise one-step M-estimate	tcov	No	beta
Local shape scatter	lcov	No	proportion, mscatter, mcdalpha, covstandard

Testing against R

Since all scatter functions are also available in several different R Packages we can conveniently test the by running the R code comparing the results. An example is given below.

```
using DataFrames
using RCall
using Test

## Load R libraries
R"library(ICSOutlier);
R"library(ICSClust);

cc=R"ICS_cov4($X, location='mean')";    # call cov4() in R
ss=cov4(X);                            # call cov4() in Julia

## Compare the returned labels, locations and scatters
@test(ss.label == rcopy(cc)[Symbol("label")])
@test(isapprox(ss.location, rcopy(cc)[Symbol("location")]))
@test(isapprox(ss.scatter, rcopy(cc)[Symbol("scatter")]))
```

Benchmarking

One of the main advantages of Julia is the so called solving of the "two languages problem", i.e. in order to achieve satisfactory computational speed in R or Python it is necessary to program critical parts of the code in a native compiled language like C or C++. This is not necessary in Julia - everything is written in the same language. A good example for this is the scatter function `tcov()` which in R is written in C++. To illustrate this we will benchmark this functions.

```
using BenchmarkTools
using ICSTools
using Random

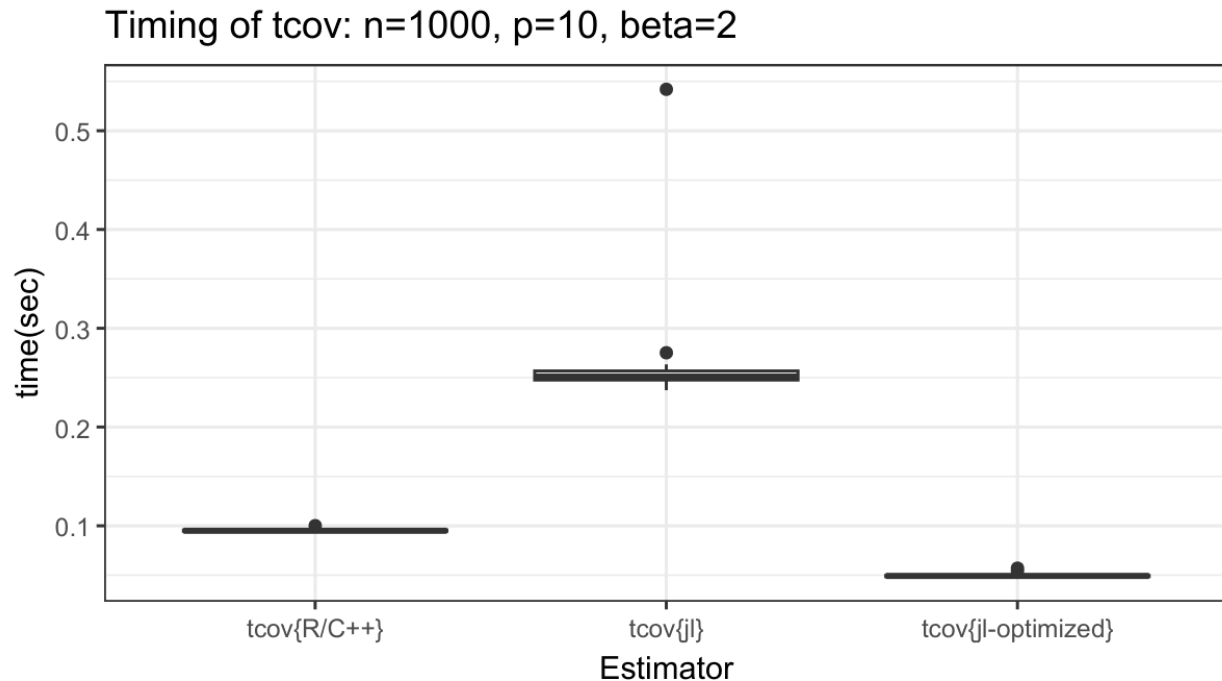
X = rand(1000, 10);

bb1 = @benchmark tcov(X) samples=19;
bb1

## BenchmarkTools.Trial: 19 samples with 1 evaluation per sample.
## Range (min ... max): 109.837 ms ... 149.193 ms GC (min ... max): 0.00% ... 0.00%
## Time (median): 117.776 ms GC (median): 0.00%
## Time (mean ± ): 119.660 ms ± 7.982 ms GC (mean ± ): 0.00% ± 0.00%
##
```

```
##
##
## 110 ms          Histogram: frequency by time          149 ms <
##
## Memory estimate: 161.86 KiB, allocs estimate: 28.
```

The following graph presents the timing of two versions of the function `tcov()` in *Julia*, the first is straight-forward translation of the C++ code in the R package *ICSClust*, the second is optimized for *Julia*.



Example applications

The Penguins data set

Data on adult penguins covering three species found on three islands in the Palmer Archipelago, Antarctica, including their size (flipper length, body mass, bill dimensions), and sex. The data set `penguins` in *R* is a data frame with 344 rows and 8 variables. It is also available in *Julia* as a package `PalmerPenguins.jl`.

Be careful there are missing values so we could consider only 342 observations. However, there are further 9 observations for which the variable `sex` has `NA`, therefore it is better to work with the remaining 333 observations.

The idea of the penguins data set is to replace the well known `iris` data set which is used throughout data science, education statistical software and machine learning as test and illustration data set. The structure of `penguins` is similar to that of `iris` but still there are several essential differences: There are several grouping variables (species, island, sex and year), the sample sizes of the groups (152-68-124) are not equal as in `iris` (3x50), there are a few observations with missing values.

Table 1. Sample sizes for `iris` and `penguins`. Data in penguins can be further grouped by island and study year.

Iris species	Sample size	Penguin species	Female	Male	NA
setosa	50	Adélie	73	73	6
versicolor	50	Chinstrap	34	34	0

Iris species	Sample size	Penguin species	Female	Male	NA
virginica	50	Gentoo	58	61	5

We start by downloading the data set from R:

```
## Load the penguins data from R
using DataFrames
using RCall
Robj = R"data('penguins', package='datasets'); x=penguins";

## Copy the contents of an R object into a corresponding canonical Julia type
penguins = rcopy(Robj);
size(penguins)
penguins = dropmissing(penguins); # drop the missing values
size(penguins)
X = penguins[:,3:6];           # select only the quantitative variables

## Tabulate species by sex
gdf = groupby(penguins, [:species, :sex]);
tt = combine(gdf, nrow);
show(IOContext(stdout, :limit=>false), MIME"text/plain"(), tt)
```

```
## 6x3 DataFrame
## Row species sex nrow
## Cat... Cat... Int64
##
## 1 Adelie female 73
## 2 Adelie male 73
## 3 Chinstrap female 34
## 4 Chinstrap male 34
## 5 Gentoo female 58
## 6 Gentoo male 61
```

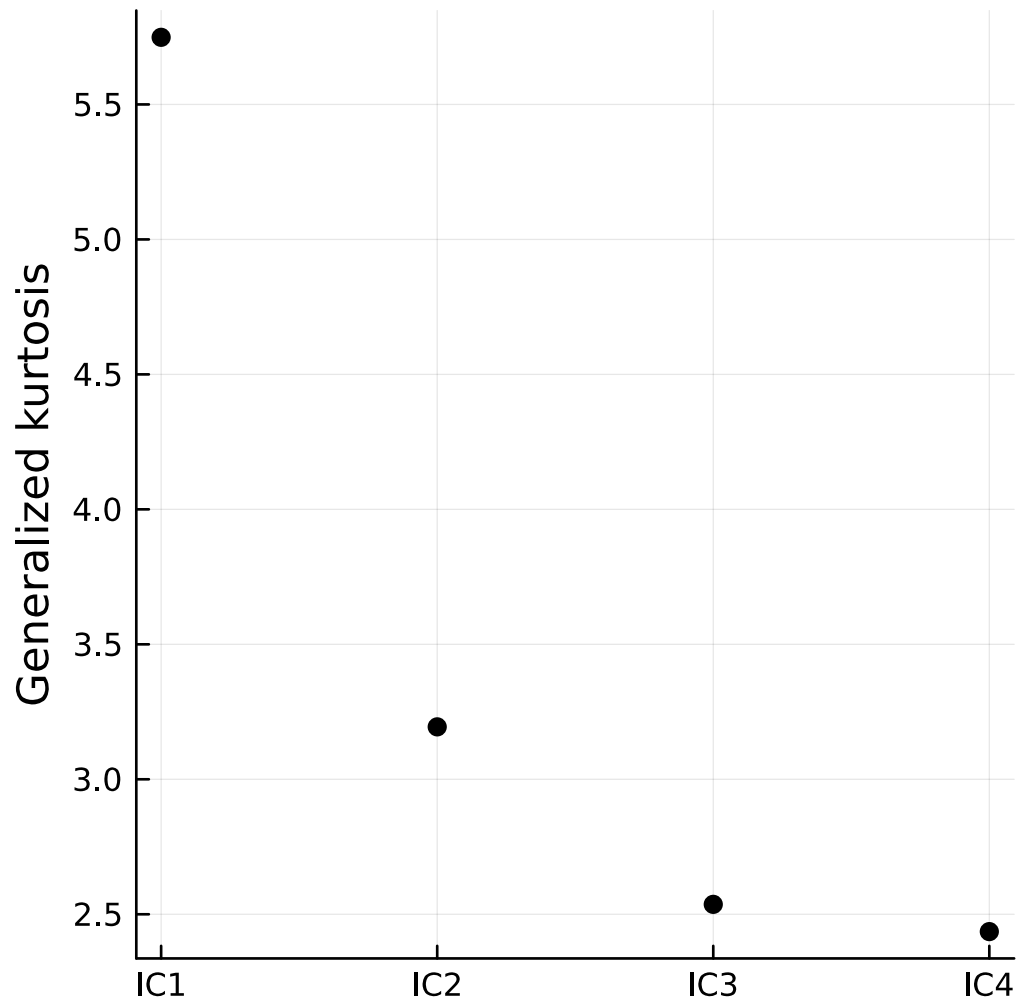
Now we apply ICS with scatters `tcov` and `cov2`, transform the model using the penguin data and plot the `scree_plot` (analogue to eigenvalues screeplot in PCA).

```
using ICSTools
ics = ICSModel(S1=tcov, S2=cov2);
scores = fit_predict!(ics, X);
```

```
using Plots
gr()
```

```
## Plots.GRBackend()
```

```
scree_plot(ics)
```

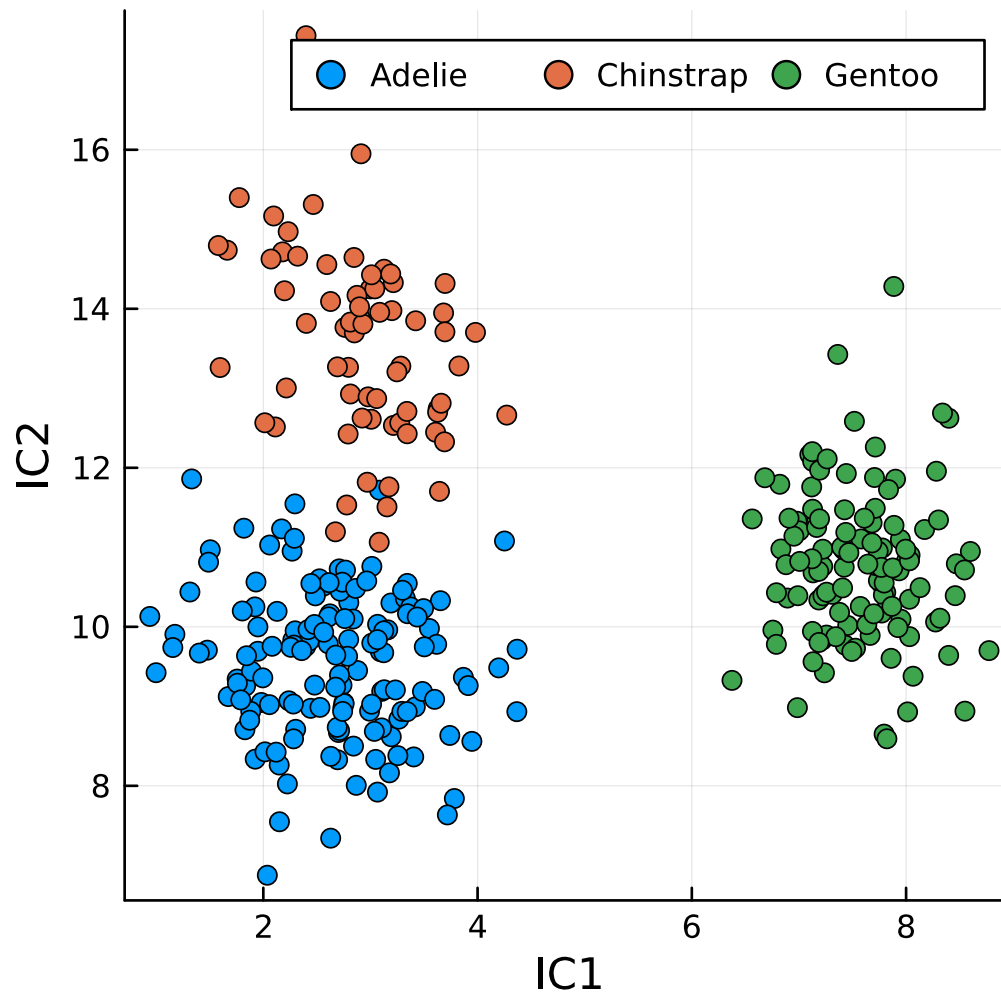


For convenience, to look at the data taking the grouping by species and by sex into account we convert the categorical arrays into string arrays.

```
## Convert the categorical arrays species and sex to string arrays  
using CategoricalArrays  
species = unwrap.(penguins.species);  
sex = unwrap.(penguins.sex);
```

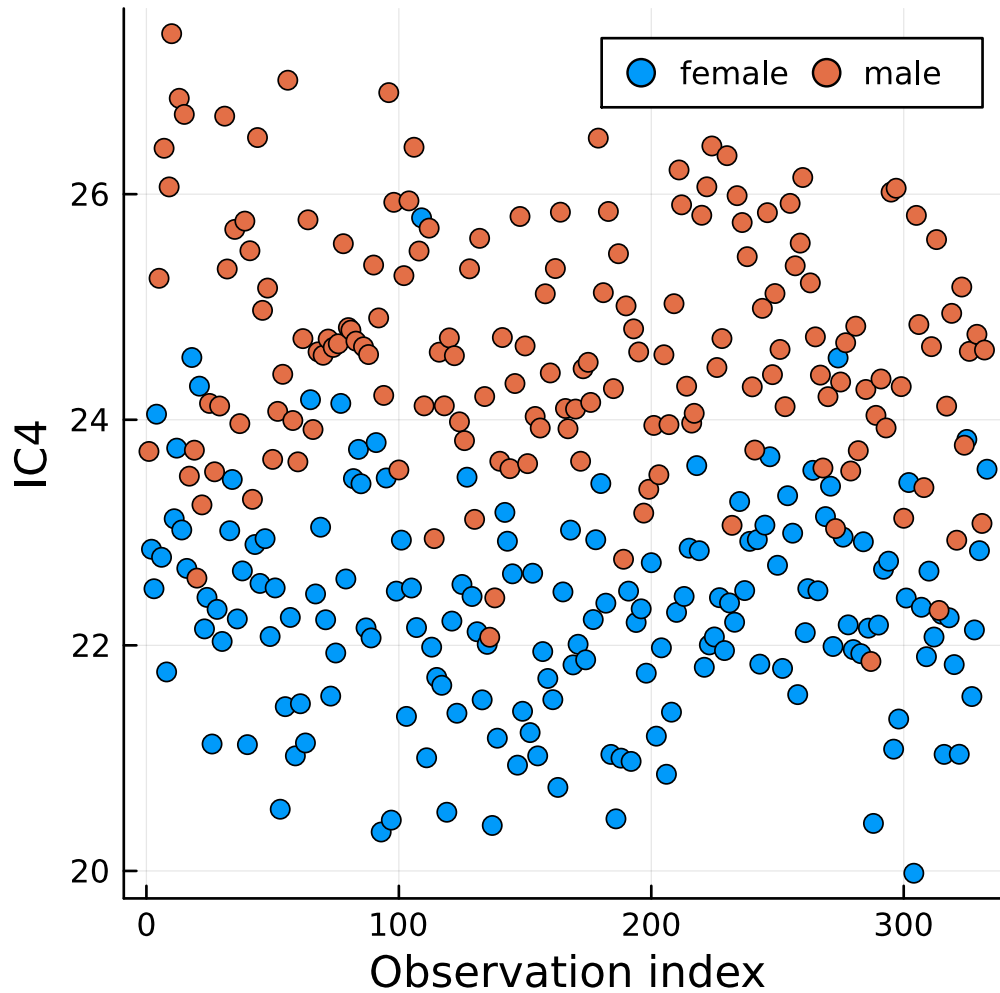
Then we plot the first two IC components grouping the data by species (the different colours represent the different species). We see that the species are very well separated.

```
## Convert the categorical arrays species and sex to string arrays  
component_plot2(ics, clusters=species)
```



Next, we plot the fourth IC component grouping the data by sex . We see that the sex of the penguins is very well separated on the fourth IC.

```
## Convert the categorical arrays species and sex to string arrays
component_plot2(ics, clusters=sex, select=[4])
```



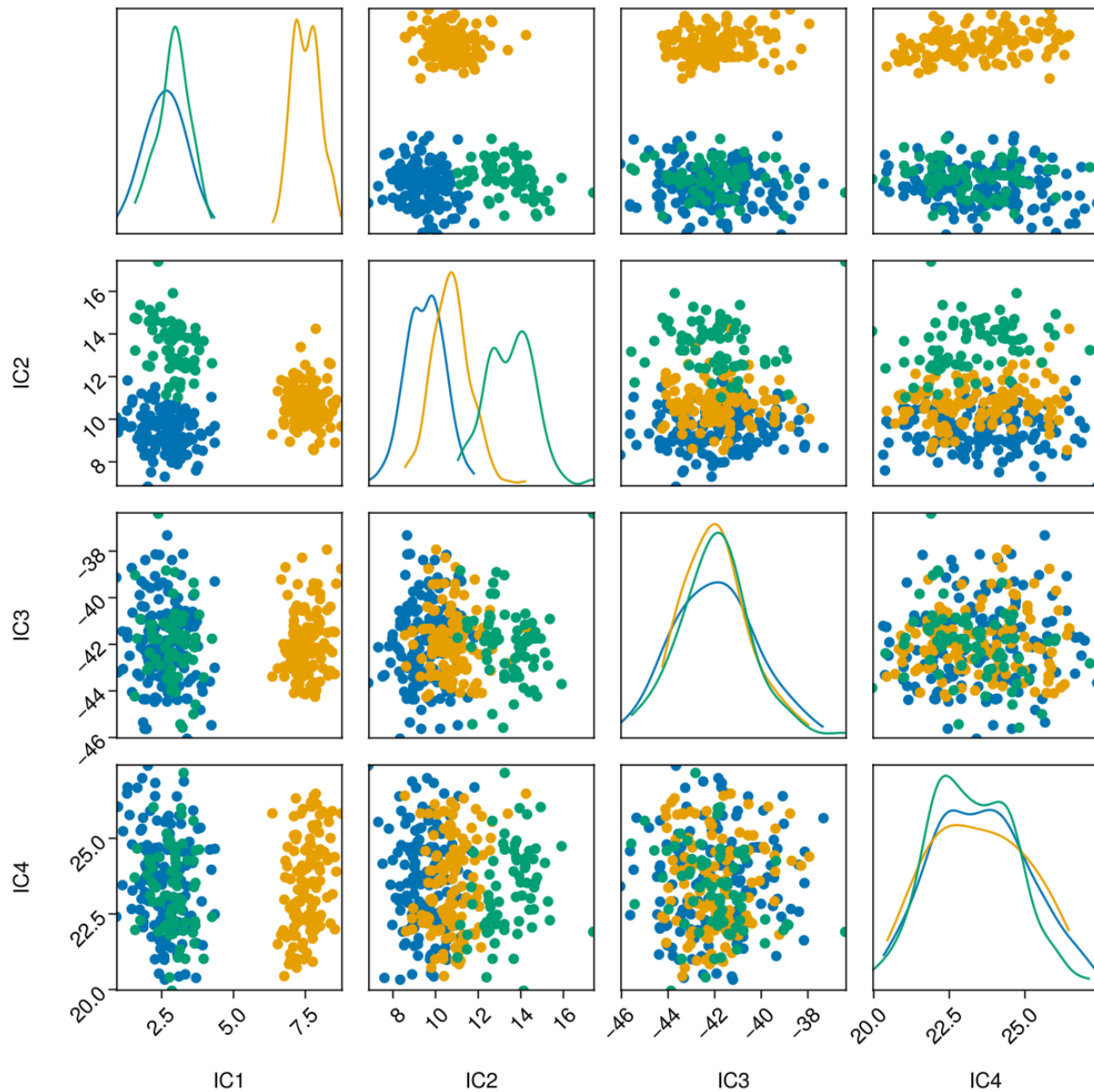
Finally, we look at the pair plots of the IC scores of the penguins data. This function (`component_plot()`) is not implemented in the `ICSTools` package, therefore we call directly the `pairplot()` function. The grouping is by species.

```
using DataFrames
using GLMakie
using PairPlots

ss = DataFrame(scores, ["IC$i" for i=1:size(scores,2)]);

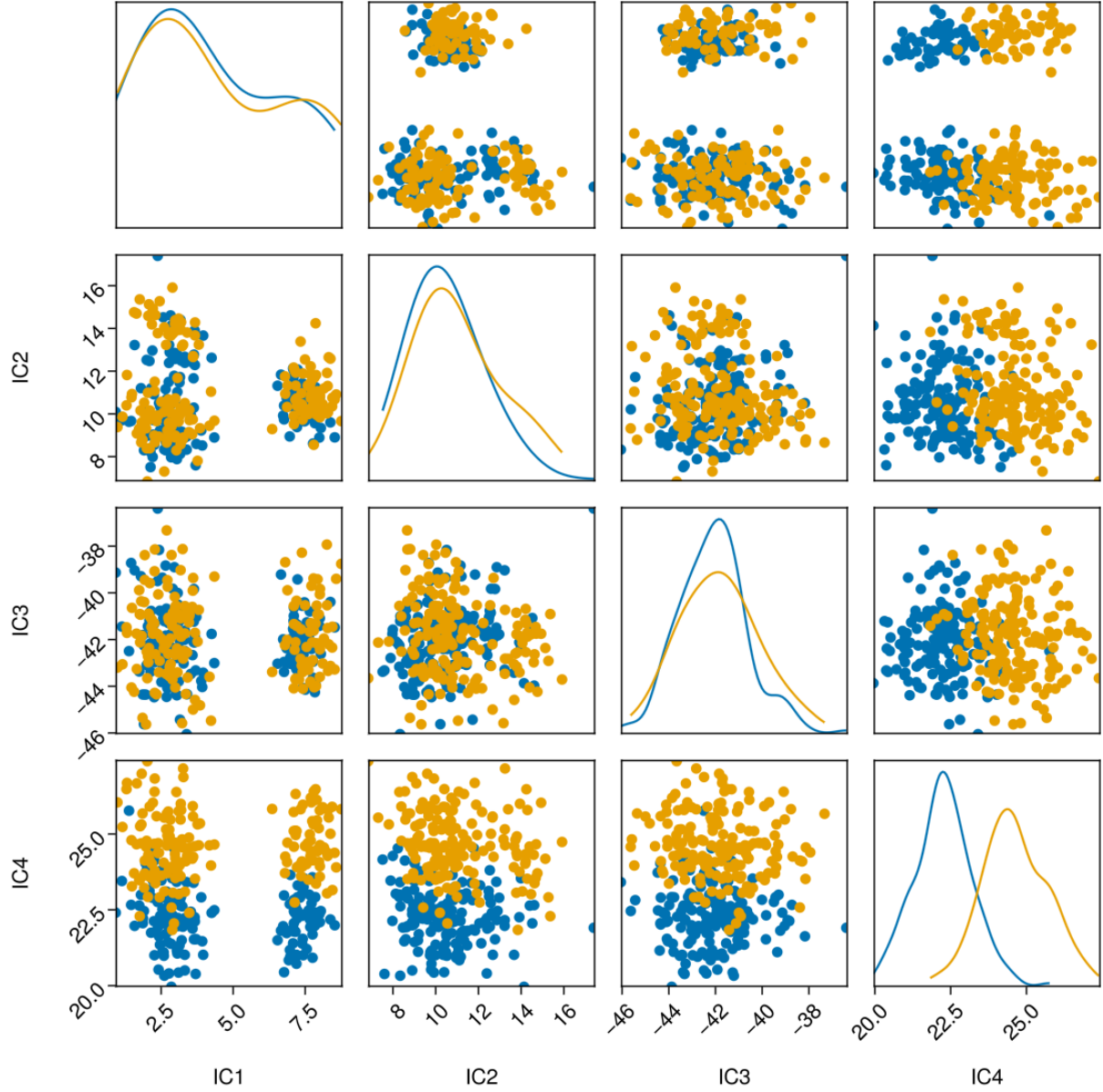
fig=pairplot(ss[species .== "Adelie", :] =>
  (PairPlots.Scatter(markersize=10),
   PairPlots.MarginDensity()),
ss[species .== "Gentoo", :] =>
  (PairPlots.Scatter(markersize=10),
   PairPlots.MarginDensity()),
ss[species .== "Chinstrap", :] =>
  (PairPlots.Scatter(markersize=10),
   PairPlots.MarginDensity()),
fullgrid=true)
```

```
save("images/penguins-plot-species.png", fig)
```



Now we repeat the same with grouping by sex.

```
fig=pairplot(ss[sex .== "female", :] =>
  (PairPlots.Scatter(markersize=10),
   PairPlots.MarginDensity()),
  ss[sex .== "male", :] =>
  (PairPlots.Scatter(markersize=10),
   PairPlots.MarginDensity()),
  fullgrid=true)
save("images/penguins-plot-sex.png", fig)
```



Cluster analysis

The Philips data set was first introduced in the robustness literature in Rousseeuw and Van Driessen (1999) as a motivation example for developing the FASTMCD algorithm. Later it was used as an example in a number of papers and presentations (Maronna and Zamar, 2002; Willems et al., 2009).

The data set is available in the R package `cellwise`. For each of $n = 677$ parts (metal plates molded by a press) $p = 9$ characteristics were measured. The aim of the multivariate analysis was to gain insight in the interrelations between the variables and to investigate if any deformations or abnormalities have occurred during the production process. There are two groups of outliers (as analyzed by the Philips engineers and demonstrated by Rousseeuw et al. (1999): the first 100 observations and the 75 observations with indexes 491 to 565. The third group are the 502 regular observations.

We start by downloading the Philips data set from R:


```

## Load the penguins data from R
using DataFrames
using RCall
RObj = R"data('data_philips', package='cellWise'); x=data_philips";

## Copy the contents of an R object into a corresponding canonical Julia type
X = rcopy(RObj)
size(X)

## Define the clusters
clusters = repeat(["Group1"], outer=size(X, 1))
clusters[1:100] .= "Group2"
clusters[491:565] .= "Group3"

## We will use these to validate the k-means
true_labels = fill(1, size(X, 1))
true_labels[1:100] .= 2
true_labels[491:565] .= 3

```

Now we apply ICS with scaters `mcd_raw` and `cov2`, transform the model using the Philips data and plot the `scree_plot` (analogue to eigenvalues scree plot in PCA).

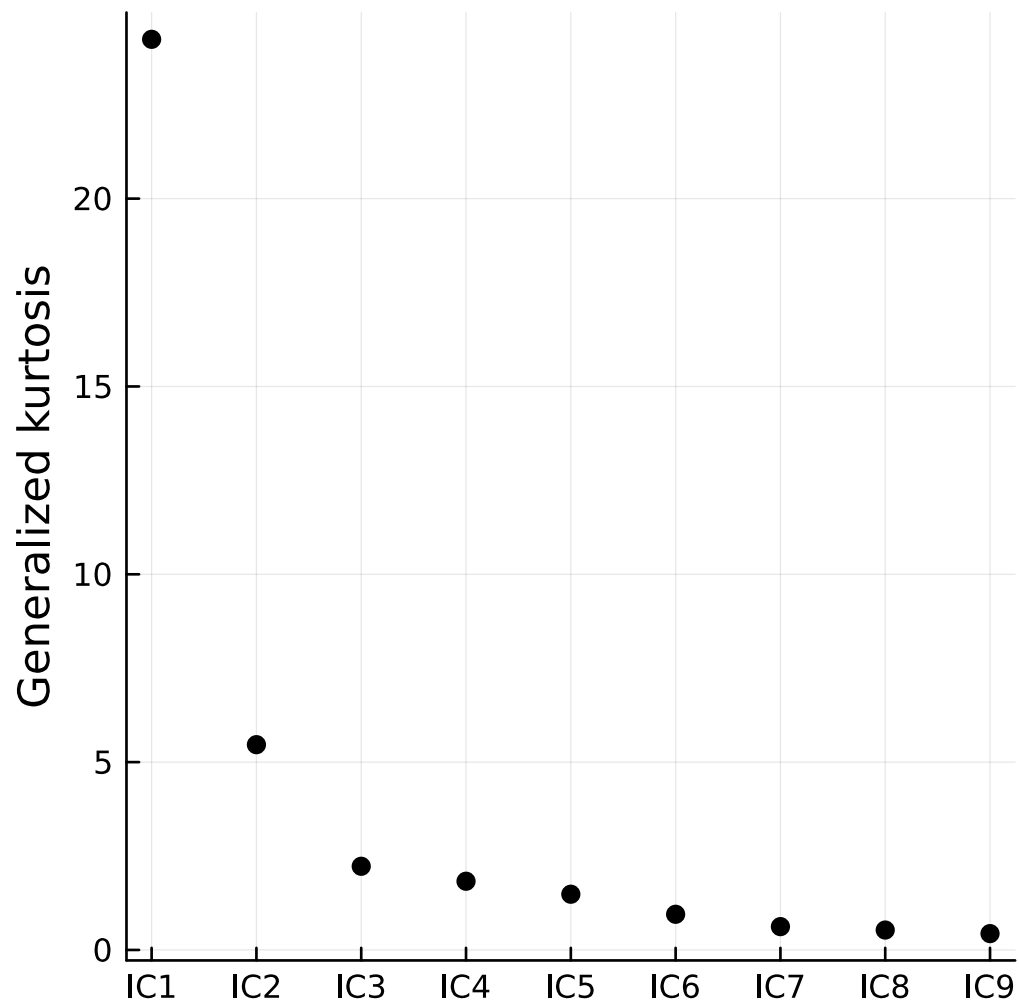
```

using ICSTools
ics = ICSModel(S1=mcd_raw, S2=cov2);
scores = fit_predict!(ics, X);

using Plots
gr()

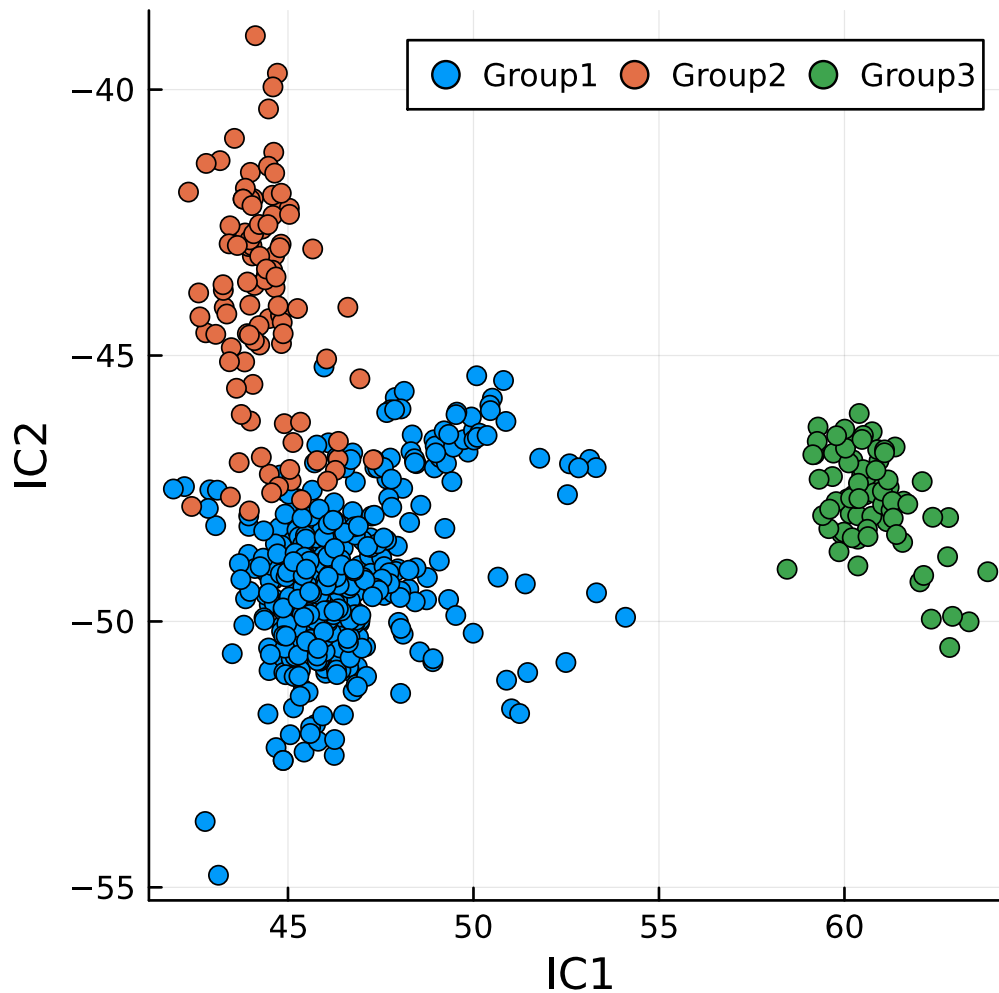
## Plots.GRBackend()
scree_plot(ics)

```



Show the clusters in the first two ICs

```
component_plot2(ics, clusters=clusters)
```

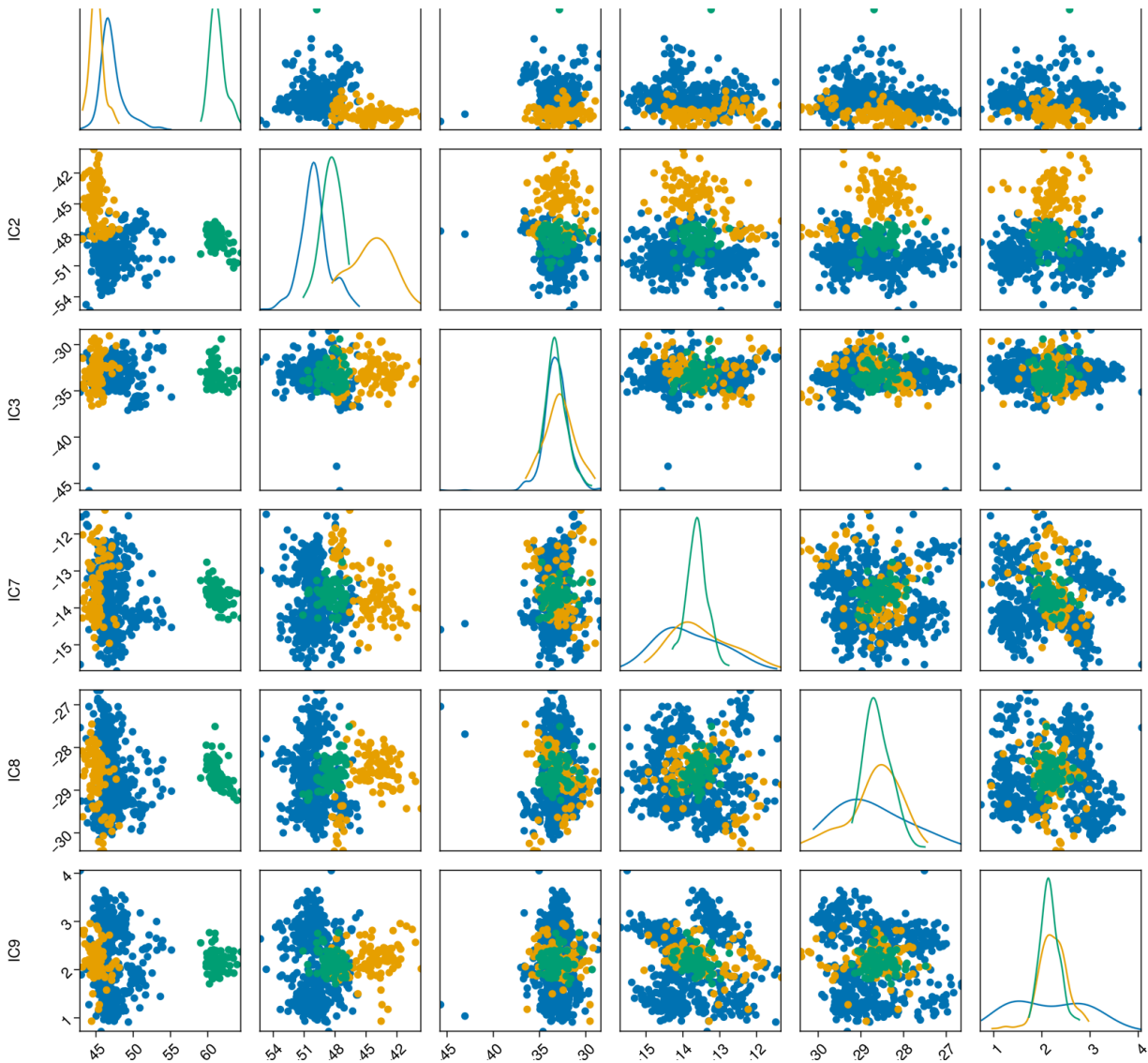


Show the pair plots of the first 3 and the last three ICs.

```
using DataFrames
using GLMakie
using PairPlots

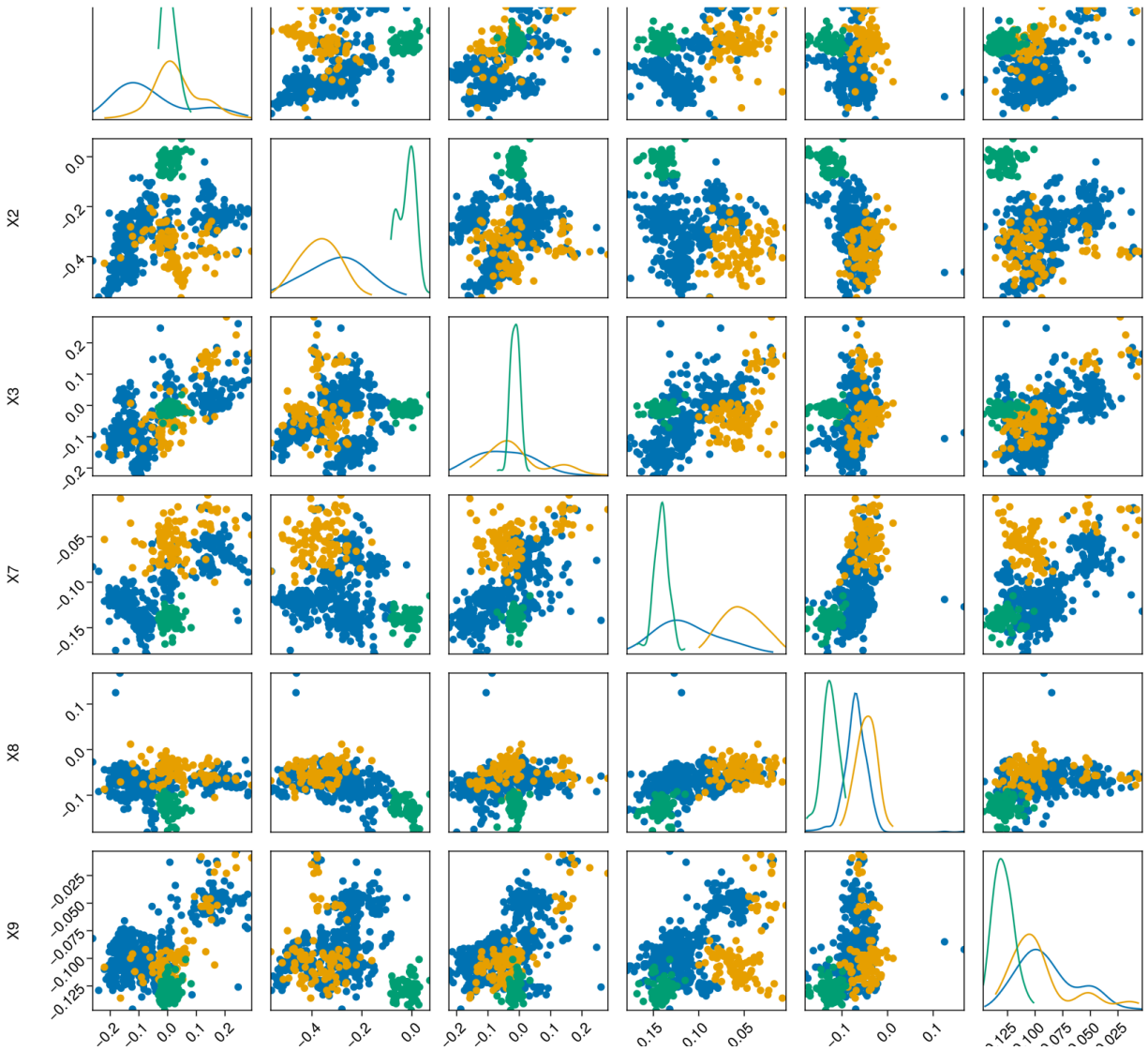
ss = DataFrame(scores[:, [1, 2, 3, 7, 8, 9]], ["IC$i" for i=[1, 2, 3, 7, 8, 9]]);
ss_raw = DataFrame(X[:, [1, 2, 3, 7, 8, 9]], ["X$i" for i=[1, 2, 3, 7, 8, 9]]);

fig=pairplot(ss[clusters .== "Group1", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
    ss[clusters .== "Group2", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
    ss[clusters .== "Group3", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
    fullgrid=true)
save("images/philips-pairs.png", fig)
```



Now show the pair plots of the original data. We cannot observe any structure.

```
fig_raw=pairplot(ss_raw[clusters .== "Group1", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
ss_raw[clusters .== "Group2", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
ss_raw[clusters .== "Group3", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
fullgrid=true)
save("images/philips-raw.png", fig_raw)
```



Next, let us look at the PCA. Again there is no visible structure.

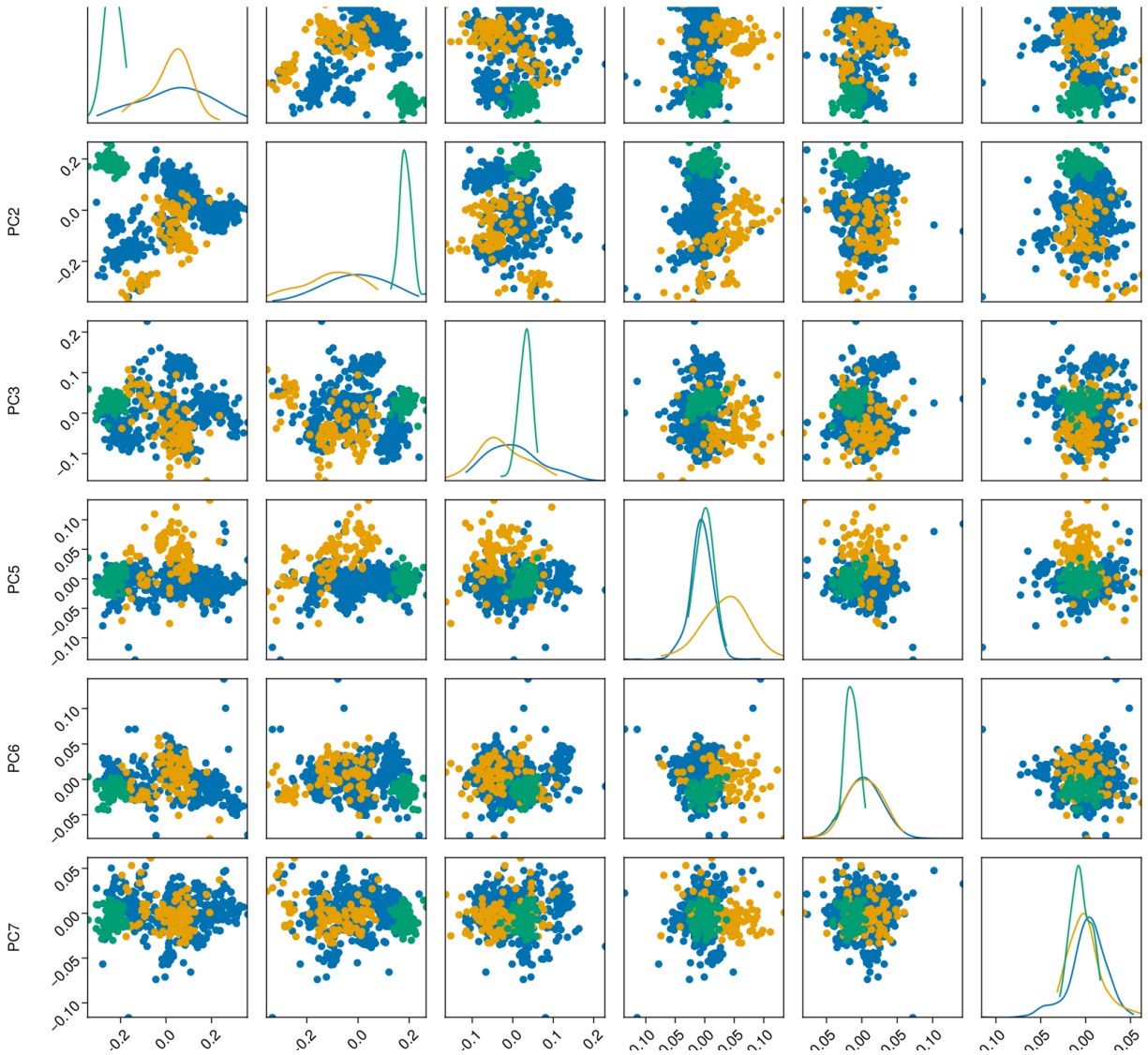
```
using MultivariateStats
Xtr=Matrix(X)'
M = fit(PCA, Xtr; maxoutdim=9)
Ytr = MultivariateStats.predict(M, Xtr)'
ss_pca = DataFrame(Ytr[:, [1, 2, 3, 5, 6, 7]], ["PC$i" for i=[1, 2, 3, 5, 6, 7]]);

fig_pca=pairplot(ss_pca[clusters .== "Group1", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
ss_pca[clusters .== "Group2", :] =>
    (PairPlots.Scatter(markersize=10),
    PairPlots.MarginDensity()),
ss_pca[clusters .== "Group3", :] =>
```

```

(PairPlots.Scatter(markersize=10),
 PairPlots.MarginDensity()),
fullgrid=true)
save("images/philips-pairs_pca.png", fig_pca)

```



Now let us turn the exploratory observations on the three pair plots to more concrete numbers. We can run k-means clustering on the raw data, the (first 3) PCA scores and the (first 3) ICs. Since we know the true clusters, we can compute a suitable external validation measure like the Adjusted Rand index and compare them.

```

## 1. Do k-means on the raw data
using Clustering
result = kmeans(X', 3);
ri, ari_raw = randindex(result.assignments, true_labels);

```

```

ari_raw

## Do k-means on the first 3 ICs
using Clustering
Y=scores[:,1:3];
result_ICS = kmeans(Y', 3);
ri, ari_ICS = randindex(result_ICS.assignments, true_labels);
ari_ICS

## Do k-means on the first 3 PCs
using MultivariateStats
using Clustering
Xtr=Matrix(X)';
M = fit(PCA, Xtr; maxoutdim=9);
Ytr = MultivariateStats.predict(M, Xtr)';
Ytr3 = Ytr[:,1:3]';
result_PCA = kmeans(Ytr3, 3);
ri, ari_PCA = randindex(result_PCA.assignments, true_labels);
ari_PCA

println("ari_raw:\t\t", ari_raw, "\nari_PCA_3:\t", ari_PCA, "\nari_ICS_3:\t", ari_ICS)

## ari_raw:      0.6107260538575162
## ari_PCA_3:    0.6115345284189734
## ari_ICS_3:    0.9442545864543365

```

Julia packages: Technical issues

Documentation

Testing

Built-in data sets